

THE ROSETTA STONE

A Necessity-Based Grammar of Physical Law

Distinction · Transformation · Constraint, and What They Do — and Do Not — Explain

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$$G = (D, T, C)$$

R emerges from (D, T, C) under sufficient constraint

PERSISTENCE IS A DERIVED QUANTITY, NOT A PRIMITIVE

The Operational Triad and the Constraint Necessity Theorem

ABSTRACT

This paper proposes a definition of “Theory of Everything” (TOE) that differs from the standard usage. The standard usage treats a TOE as a single physical theory unifying the four fundamental forces under one Lagrangian. Under that definition, no TOE exists, and the search has stalled for half a century. This paper argues that the standard definition conflates two distinct projects: finding the deepest object reality is made of, and finding the deepest necessity that any stable, describable reality must satisfy. We adopt the second project. We define a Theory of Everything as the minimal explanatory grammar shared by every mature, predictively successful physical theory — the structural architecture that recurs without exception once domain-specific vocabulary is removed. We show that this architecture is a three-primitive grammar, Distinction (D), Transformation (T), and Constraint (C), from which Retention/Persistence (R) is derived rather than assumed. We demonstrate the grammar's instantiation across six established physical and formal domains, present a rigorous derivation showing that the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is the minimal anomaly-free structure satisfying local identity preservation under scale reduction, ground the convergence mechanism in the Banach Fixed-Point Theorem, and state falsifiability conditions and open problems with the same rigor used elsewhere in physics. We do not claim a new physical force, particle, or equation. We claim that the equipment for a Theory of Everything, under this corrected definition, already exists inside established twentieth- and twenty-first-century physics and mathematics — distributed across general relativity, quantum field theory, information theory, and dynamical systems theory — and that what

has been missing is the recognition that these are not separate theories awaiting unification by a new force, but independent confirmations of a single underlying grammar.

Keywords: Theory of Everything, Necessity Principle, Distinction-Transformation-Constraint, Renormalization Group, Banach Fixed-Point Theorem, Gauge Anomaly Cancellation, Universality Audit, Structural Equivalence

1. A CORRECTED DEFINITION OF THEORY OF EVERYTHING

The search for a Theory of Everything in twentieth- and twenty-first-century physics has operated under a specific, mostly unstated definition: a TOE is a single field theory, with a single Lagrangian, that reduces general relativity and the Standard Model to special cases of one deeper structure. String theory, loop quantum gravity, and asymptotic safety are each attempts to satisfy this definition. None has succeeded in a way the physics community regards as settled, and the search has now continued for longer than the gap between Newton and Einstein.

This paper argues that the stalled search is evidence about the definition, not only about the difficulty of the mathematics. The standard definition assumes that explanation terminates at an object: find the right field, the right string, the right loop, and the unification follows. But the history of physics does not actually run this way. Einstein did not begin general relativity by proposing a new substance. He imposed requirements — locality, covariance, conservation — and the geometry of spacetime followed as the structure that satisfies them. Noether did not begin with conservation laws as a brute fact; she imposed symmetry requirements, and conservation followed as a theorem. Shannon did not begin with entropy as a postulate; he began with the requirement of distinguishable messages, and entropy followed as the unique measure consistent with that requirement.

In each historical case, necessity preceded structure. The equation was the output of a constraint-satisfaction problem, not the input. This paper takes that pattern seriously as a methodological principle and applies it to the TOE question itself. Instead of asking what reality is ultimately made of, we ask: what constraints must be satisfied for any stable, distinguishable, describable structure to exist and to be investigated at all? A theory that answers this question is a Theory of Everything in the corrected sense — not because it specifies the content of every physical law, but because it specifies the architecture that every successful physical law, regardless of content, has been shown to share.

This is a narrower and more defensible claim than the standard TOE ambition, and it is important to be explicit about what it does and does not deliver. It does not predict the mass of the electron, the value of the cosmological constant, or a new force. It claims that the explanatory architecture of physical law — distinguishable states, admissible transformations between them, and constraints selecting which transformations occur — is itself the deepest layer currently reachable, and that this architecture is what every fundamental theory in physics has independently rediscovered under different names.

2. THE OPERATIONAL TRIAD: $G = (D, T, C)$

The grammar rests on three primitives, each given a formal definition, an intuitive gloss, and an operational test — a specification of what a researcher would look for in a given theory to determine whether the primitive is instantiated.

2.1 D — DISTINCTION

$$D : S \rightarrow \{0,1\}; \quad D(s) = 1 \quad \text{iff} \quad s \text{ is distinguishable from at least one } s' \in S$$

Formal definition

Distinction is the precondition for a state to be a determinate entity at all. Operationally, D is identified by the existence of a measurement, observable, or classification rule that assigns different values to different states. A theory with no such criterion has not specified a subject matter.

2.2 T — TRANSFORMATION

$$T : S \times S \rightarrow [0,1]; \quad T(s, s') = \text{probability of transition from } s \text{ to } s'$$

Formal definition

Transformation characterizes how states change, evolve, or are causally produced from one another. Operationally, T is identified by the existence of dynamics, equations of motion, or causal/morphic structure carrying states from one configuration to another. Domains with no T — pure mathematics is the clearest case — are investigated by proof rather than experiment; the framework predicts this distinction rather than being embarrassed by it.

2.3 C — CONSTRAINT

$$C \subseteq T; \quad C = \{t \in T \mid t \text{ is admissible under the theory's laws}\}$$

Formal definition

Constraint restricts which transformations actually occur. Operationally, C is identified by symmetry conditions, conservation laws, selection rules, boundary conditions, or cost functions. The Constraint Necessity Theorem, proven below, establishes that constraint is not merely a restriction on persistence — it is the condition that makes persistence possible at all.

2.4 R — RETENTION, AS A DERIVED QUANTITY

$$R(s, t_0, t_1) = 1 \quad \text{iff} \quad D(s) \text{ is preserved under } T|C \text{ from } t_0 \text{ to } t_1$$

Formal definition

This is the central architectural correction relative to earlier formulations of this program. Persistence is not posited as a fourth primitive. It is derived from the other three. A structure persists because it survives the constrained transformation — not because persistence was separately assumed to exist. The two theorems below establish this derivation rigorously.

THEOREM 1 – Constraint Necessity Theorem
 $(D, T, \neg C) \rightarrow \neg P$

If the constraint set C is empty, persistent distinguishable structure P cannot be guaranteed: the system relaxes to a maximum-entropy distribution over all reachable states, and no particular state is retained above chance.

Proven by direct argument from the definition of T and C ; see §6.1

THEOREM 2 – Retention Emergence
 $(D, T, C) \rightarrow R \quad (\text{under sufficient } C)$

Given a non-empty proper constraint set, the states stable under C – the fixed points or attractors of the constrained dynamics – constitute R . Follows from the Brouwer Fixed-Point Theorem (continuous case) and its discrete analogs.

Established result; see §6.1

Theorem 1 and Theorem 2 together are the formal core of the entire framework. They are not metaphors. Theorem 1 is a direct consequence of the definition of an unconstrained Markov-type transition structure: with all transitions equally admissible, the system's state distribution converges to maximum entropy, and the probability of recovering any particular initial state vanishes. Theorem 2 invokes a standard fixed-point result: a continuous self-map of a compact convex set has a fixed point, and that fixed point is exactly what a constrained dynamics stabilizes onto. The contribution of this paper is not these theorems individually — they are standard mathematics — but the demonstration, in Section 4, that this exact pattern recurs without modification across six independently developed branches of physics and formal science.

3. WHY “REPLACEMENT” WAS THE WRONG SEARCH STRATEGY

Every major TOE program of the last fifty years — string theory, loop quantum gravity, asymptotic safety — has been organized around an implicit assumption: that general relativity and the Standard Model are incomplete theories awaiting replacement by a deeper one. This paper argues that the assumption itself is the obstacle, and that it is worth stating precisely why.

General relativity and quantum field theory are not arbitrary historical accidents that happen to disagree at the Planck scale. Both are highly constrained solutions to specific necessity problems. General relativity is the structure forced by requiring locality, general covariance, and conservation of stress-energy in a theory of gravitation; alternative theories satisfying the same constraints (such as scalar-tensor gravity) exist but require additional fields not demanded by the constraints themselves, which is why general relativity remains the minimal solution. Quantum field theory is the structure forced by requiring unitary, causal, Lorentz-invariant evolution of distinguishable quantum states. Section 5 below shows, using only established quantum field theory, that the specific gauge content of the Standard Model is itself forced — not chosen — once a small number of necessity conditions are imposed.

If both theories are necessity solutions rather than approximations of a deeper substance, then the failure to find a single replacing object is not evidence of insufficient cleverness. It is evidence that the question “what replaces them?” was malformed from the outset. The corrected question is: what necessity conditions force structures of exactly this kind to appear, and is that necessity architecture itself the deepest available answer to “what is reality?”

4. THE CROSS-DOMAIN AUDIT: ESTABLISHED PHYSICS ONLY

Table 1 instantiates $G = (D, T, C, R)$ in six domains, each governed by a mature, independently developed, experimentally validated theory. Nothing in this table is proposed by this paper. Each entry is the standard formal content of the cited theory. What is proposed is only the observation that the same four structural roles recur, without exception, across all six — and that no sub-discipline of physics or formal science has yet been found that lacks any of the three primitives.

Domain	D — Distinction	T — Transformation	C — Constraint	R — Retention
General Relativity	Spacetime events (4-manifold)	Geodesics, diffeomorphisms	Einstein field equations	Stress-energy conservation
Quantum Mechanics	Hilbert space vectors	Unitary evolution, measurement	Schrödinger eq., superselection	Probability, quantum numbers
Thermodynamics	Macrostates (T,P,V,S)	Thermodynamic processes	First & Second Laws	Equilibrium, entropy monotonicity

Domain	D — Distinction	T — Transformation	C — Constraint	R — Retention
Information Theory	Symbols, messages	Encoding, transmission	Channel capacity bound	Recoverable messages
Evolutionary Biology	Genotypes / phenotypes	Mutation, drift, recombination	Fitness landscape	Adaptive trait persistence
Control Theory	State-space vectors	System dynamics	Lyapunov stability conditions	Stable equilibria

Table 1. The Cross-Domain Structural Audit. Each row is the standard formal content of an independently established theory; the D/T/C/R columns are a re-description, not a new claim about the physics itself. The claim under audit is that the architecture, not the content, is invariant.

This table should be read adversarially. A reasonable objection is that any sufficiently flexible vocabulary can be made to fit any theory — that “state space,” “dynamics,” and “constraint” are general enough to describe almost anything, which would make the table trivially true rather than informative. Section 6.4 addresses this objection directly under the heading of over-compression risk, and states the specific test — the Structural Equivalence Test — that distinguishes a genuine architectural finding from a vocabulary coincidence.

5. THE GAUGE GROUP AS A FORCED STRUCTURE, NOT A POSTULATE

The strongest evidence available for the necessity framework is a derivation, not an analogy. This section shows that the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ can be obtained as the unique minimal structure satisfying three independent, well-established physical requirements, rather than being introduced as a postulate of the Standard Model Lagrangian.

5.1 LOCAL PHASE INVARIANCE FORCES A GAUGE CONNECTION

Let $\Psi(x)$ be a local quantum state description. Demanding that the overall phase of Ψ be a matter of convention at every point independently — a local $U(1)$ phase freedom $\Psi(x) \rightarrow e^{i\theta(x)}\Psi(x)$ — is incompatible with the ordinary derivative, because $\partial_\mu \Psi$ picks up an unwanted term proportional to $\partial_\mu \theta(x)$. This is standard, textbook quantum field theory, not a novel claim. The unique repair is the introduction of a compensating gauge connection $A_\mu(x)$, forming the covariant derivative:

$$D_\mu = \partial_\mu + i A_\mu(x), \quad A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu \theta(x)$$

The covariant derivative — standard QFT result

This is the textbook derivation of why gauge fields exist at all: they are not a free choice but an algebraic necessity for maintaining local phase consistency. The contribution of this paper is only the framing — reading this as an instance of the Constraint Necessity Theorem: a constraint (local identity preservation) forces a transformation structure (the gauge connection) into existence.

5.2 CHIRALITY AND ANOMALY CANCELLATION FORCE THE SPECIFIC GROUP

Three independent, well-established constraints narrow the admissible gauge group:

- (i) **Local phase preservation** requires at minimum an abelian $U(1)$ connection, as shown above, but $U(1)$ alone has no internal degrees of freedom to support a non-abelian self-interaction or to split chiral (left/right-handed) representations — a $U(1)$ -only theory cannot generate massive, chirally asymmetric matter.
- (ii) **Chiral asymmetry** the observed fact that weak interactions act only on left-handed fermion fields requires the minimal non-abelian extension $SU(2)$. A standalone $SU(2)$ gauge theory with an odd number of fermion doublets, however, suffers the Witten global anomaly: the path integral changes sign under a large gauge transformation, $Z \rightarrow -Z$, forcing $Z \equiv 0$ identically — a mathematically inconsistent theory. This is an established, peer-reviewed result (Witten, 1982).
- (iii) **Triangle anomaly cancellation** even after fixing $SU(2)_0 \times U(1)_Y$, the chiral axial-vector triangle diagram $\text{Tr}[\{T^a, T^b\} T^c]$ does not vanish for the electroweak sector alone. Cancelling it without hand-assigning charges requires an additional non-abelian structure; the minimal group that achieves exact cancellation, with the observed hypercharges, is $SU(3)$ at exactly $N_c = 3$ colors — verified by

the identity $\Sigma Q = 3(2/3 - 1/3) + (-1) = 0$. This is the standard anomaly-cancellation argument used throughout quantum field theory pedagogy (see Peskin & Schroeder, 1995, ch. 19–20), not a new computation.

$$G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$$

the minimal anomaly-free group satisfying (i)–(iii) simultaneously
Forced, not postulated

This derivation is genuinely strong evidence for the necessity framework because it shows a real, falsifiable structural fact: any group smaller than G_{SM} fails one of the three constraints, and the standard counter-candidates from Grand Unified Theories ($\text{SU}(5)$, $\text{SO}(10)$) are larger anomaly-free groups that satisfy the constraints non-minimally, introducing additional gauge bosons (such as those mediating proton decay) that are not forced by the necessity conditions themselves and are constrained to be extremely heavy or absent by experiment. The Standard Model gauge group is not an arbitrary list assembled from historical accident; it is the lowest-order solution to a well-posed constraint problem, and that fact was established by the ordinary methods of quantum field theory decades before this paper, independent of any necessity framework.

6. THE CONVERGENCE MECHANISM: RENORMALIZATION AS A CONTRACTION MAPPING

6.1 WHY CONSTRAINED REDUCTION CONVERGES TO A UNIQUE ATTRACTOR

The Renormalization Group (RG) flow of quantum field theory is the best-established physical instance of a constrained reduction operator R acting repeatedly on a space of descriptions. Let L be the space of admissible quantum field descriptions (couplings, field content, and symmetry data) at a given energy scale, equipped with a distance measure d between two descriptions. The physical content of the RG flow is that integrating out short-distance degrees of freedom is a dissipative, non-unitary, information-discarding operation: distinguishing features that depend on microscopic detail are washed out as the scale is lowered.

This is the same structural content as the Zamolodchikov C-theorem in two-dimensional quantum field theory (Zamolodchikov, 1986), which proves rigorously that a specific measure of degrees of freedom decreases monotonically along RG flow toward the infrared. Where the C-theorem is established for two dimensions and generalized with weaker results in higher dimensions (the a-theorem; Komargodski & Schwimmer, 2011), the general structural claim — that RG flow is information-losing and drives distinct microscopic theories toward shared macroscopic (infrared) behavior — is uncontroversial and foundational to all of modern quantum field theory.

$$d(R(\ell_1), R(\ell_2)) \leq c \cdot d(\ell_1, \ell_2), \quad c \in [0,1)$$

If (L, d) is complete and R satisfies this contraction property, the Banach Fixed-Point Theorem guarantees:

$$\lim_{n \rightarrow \infty} R^n(X) = \Delta^* \quad \text{for every starting description } X \in L$$

Banach Fixed-Point Theorem applied to the RG reduction operator

The existence and uniqueness of an infrared fixed point Δ^* under RG flow is, again, not a new claim of this paper — it is the established content of the theory of renormalization-group fixed points, used throughout condensed matter physics (universality classes of phase transitions) and in the Asymptotic Safety program for quantum gravity (Reuter & Saueressig, 2012), where Γ^* denotes the specific UV fixed point of the gravitational coupling flow. What this paper adds is the explicit recognition that this is a special case of the general Constraint-Necessity / Retention-Emergence pattern of Theorems 1 and 2: a contractive constraint operator acting on a complete space of descriptions necessarily yields a unique retained structure, regardless of the specific physical content of the description space.

6.2 SPECTRAL DIMENSION: AN INDEPENDENTLY CONFIRMED PREDICTION PATTERN

A second, independent line of established physics exhibits the identical convergence pattern. The heat-kernel trace of a Laplace-type operator on a manifold of spectral dimension D_s takes the asymptotic form:

$$P(\sigma) = \text{Tr}(e^{-\sigma \Delta}) \sim (4\pi\sigma)^{-D_s/2} \cdot \sum_m a_m(\Delta), \quad \sigma \rightarrow 0$$

$$D_s = -2 \cdot \lim_{\sigma \rightarrow 0} [\ln P(\sigma) / \ln \sigma]$$

Heat kernel trace and spectral dimension — established differential geometry

Independent numerical work in Causal Dynamical Triangulations (Ambjørn, Jurkiewicz & Loll, 2005) and in Asymptotic Safety computations has found that the spectral dimension of spacetime, computed this way, flows from $D_s \approx 4$ at macroscopic scales toward $D_s \approx 2$ near the Planck scale. This is a genuinely independent confirmation — obtained through entirely different mathematical machinery (lattice path-integral simulation rather than gauge-anomaly algebra) — of the same qualitative pattern found in Section 5: a constrained reduction process driving a system toward a lower-dimensional, structurally simpler invariant as resolution decreases. The convergence is suggestive, not proof of a single shared mechanism; the honest statement is that two unrelated established physics programs each exhibit dimensional reduction under scale-coarsening, which is consistent with, but does not by itself establish, the general necessity grammar.

7. HOSTILE TESTING

A framework that claims to identify a universal explanatory grammar is obligated to seek out its own strongest counterexamples rather than its best illustrations. The following tests were run adversarially against the (D, T, C) grammar.

7.1 SEARCH FOR A THEORY LACKING D, T, OR C

No mature physical theory was found lacking any of the three primitives. The nearest candidate, pure mathematics (number theory, geometry, formal logic), lacks T in the physical sense — the integers do not evolve — but this is a confirmation of the framework's boundary conditions rather than a counterexample: pure mathematics is investigated by proof, not experiment, and the framework correctly predicts that T-absent domains are non-empirical. A more careful look shows mathematics retains a deductive T (valid inference as transformation between formal structures), so the case is a qualification, not a failure.

7.2 SEARCH FOR A SMALLER GRAMMAR

Eliminating D leaves T and C with no domain to act on. Eliminating T leaves C with nothing to restrict, collapsing the framework to a static taxonomy with no account of change. Eliminating C, by Theorem 1, eliminates the guarantee of persistence — attempts to derive retained structure from D and T alone either fail or smuggle a constraint in through an unacknowledged restriction on T. No two-primitive reduction was found to succeed without one of these failures.

7.3 DOCUMENTED, UNRESOLVED WEAKNESSES

Honesty about a framework's limitations is part of making it scientifically credible, and three weaknesses remain genuinely open rather than resolved by rhetorical defense.

Observer dependence. The identification of D, T, and C in a given theory is performed by a human researcher using human cognitive categories. The response — that mature, adversarially-tested theories produce stable D/T/C identifications across independent observers — does not fully close the objection, because the adversarial testing is itself performed by observers of the same cognitive type.

Over-compression risk. Sufficiently loose vocabulary can be made to fit almost any structure, which would make Table 1 true by construction rather than informative. The Structural Equivalence Test (§8) is the proposed defense: it requires not just that D/T/C roles can be labeled in two theories, but that valid inferences in one theory translate to valid inferences in the other under an explicit correspondence map. A label-only match without inferential transfer does not count as structural equivalence.

Survivorship bias. The cross-domain audit examines theories that survived to maturity. It is possible that non-(D,T,C)-structured theories exist but were filtered out long before reaching the textbook

stage, for reasons that are sociological rather than structural. This objection cannot be fully answered from inside the framework; it is recorded here as an honest, unresolved limitation rather than dismissed.

8. THE STRUCTURAL EQUIVALENCE TEST

To prevent the over-compression failure mode identified above, structural equivalence between two theories X and Y is defined by a falsifiable test rather than by vocabulary matching.

$$X \equiv_s Y \quad \text{iff} \quad \exists \text{ a correspondence map } \varphi: \text{RS}(X) \rightarrow \text{RS}(Y)$$

such that every valid inference I in X translates, under φ , to a valid inference in Y

$$\text{RS}(X) = (D_X, T_X, C_X, R_X) \quad - \quad \text{the Rosetta Signature of theory } X$$

The Structural Equivalence Test

Two theories pass the test only if insight actually transfers under the mapping — not merely if both can be described using the words “state,” “dynamics,” and “constraint.” This is the same standard, applied formally, that already operates informally whenever a result from one field of physics (say, the renormalization group) is successfully imported into a structurally unrelated field (say, condensed matter universality classes or, more speculatively, ecological population dynamics). The test gives a way to distinguish a genuine import from a coincidental metaphor: does it generate new, checkable predictions in the target domain, or only a redescription of what was already known there.

9. FALSIFIABILITY

A framework of this scope must expose concrete ways it can fail. Table 2 lists conditions under which specific, identified claims of this paper — not the broad philosophical orientation, which is not itself falsifiable in the Popperian sense — would be considered refuted.

If This Is Demonstrated:	Then:
A mature, predictively successful physical theory is found that has no identifiable state space, no dynamics, and no constraint structure (no D, T, or C role instantiated)	The Minimal Grammar Conjecture is falsified
A smaller grammar (two primitives, or one) is shown to generate every cross-domain structural equivalence currently attributed to (D, T, C)	The Minimal Grammar Conjecture is falsified by compression
The anomaly-cancellation argument for $SU(3) \times SU(2) \times U(1)$ as the minimal admissible gauge group is shown to be mathematically invalid, or a smaller anomaly-free group is shown to satisfy the same boundary conditions	The gauge-derivation claim in Section 5 is falsified
The contraction property of the renormalization-style reduction operator R fails to hold (the distance between two distinct description-languages does not shrink under R)	The Banach fixed-point argument for Δ^* fails; uniqueness of the attractor is not guaranteed

Table 2. Falsifiability conditions for the specific, checkable claims in this paper. The broad necessity-based reorientation of §1–3 is a methodological proposal, evaluated by explanatory scope and internal consistency rather than by direct falsification; the items in this table are the load-bearing technical claims that can fail outright.

10. OPEN PROBLEMS

The following are stated as open, not as solved, in keeping with the standard this paper holds itself to elsewhere.

- 1. Derive the transition from Relation to Distinction.** What forces a relational structure to generate a stable distinction, rather than remaining symmetric and undifferentiated? Symmetry-breaking is the strongest existing partial mechanism but is circular as a full derivation, since defining a symmetry group already requires specifying what may be transformed — which presupposes distinguishable states.
- 2. Formalize the Minimal Grammar Conjecture.** That (D, T, C) is the unique minimal grammar generating persistent distinguishable structure is well supported by the audits in this paper but not formally proved. A proof would need to show, for an arbitrary candidate two-primitive grammar, that it cannot generate the full space of cross-domain structural equivalences identified here.
- 3. Quantitative derivation of M^* -reconstruction scaling.** The qualitative claim that systems with more alternative stable configurations recover faster from perturbation is supported across multiple domain instances (metabolic flux degeneracy, quantum error-correction code distance, neural circuit redundancy) but lacks a first-principles quantitative derivation of the scaling exponent in any single, well-specified domain.

4. Derive the dimensionless constants. Section 5 derives the gauge group structure of the Standard Model. It does not derive the fine-structure constant, the mass hierarchy, or any other dimensionless parameter. Whether these are further necessity-forced quantities or genuinely free parameters of this universe's particular solution is unresolved.

11. CONCLUSION

The claim of this paper is deliberately narrower than the claim of a standard Theory of Everything, and that narrowness is the point. We have not proposed a new force, a new particle, or a new fundamental equation. We have argued that the equipment required for a Theory of Everything, properly defined, is already distributed across established twentieth- and twenty-first-century physics: the gauge-anomaly cancellation arguments of quantum field theory, the fixed-point structure of renormalization-group flow, the Banach Fixed-Point Theorem, Noether's theorem, and Shannon's information theory. None of these needed to be invented for this paper. What was missing was the recognition that they are independent confirmations of one shared architecture — distinguishable states, admissible transformations, and constraints selecting which transformations occur — rather than separate, unrelated achievements awaiting unification by some further undiscovered object.

Under the standard definition of a TOE — a single replacing field theory — this paper offers nothing. Under the corrected definition proposed in Section 1 — the minimal necessity grammar shared by every mature physical theory — this paper argues that the search is substantially complete, that what remains is formalization (Section 10's open problems) rather than discovery of new physics, and that the half-century stall in TOE research is evidence that physics has been searching for the wrong kind of object rather than evidence that the answer lies beyond current reach.

The corrected question this paper proposes is not “what is reality made of?” It is: what must be true of any structure for it to be distinguishable, to change, and to persist under constraint? The answer, supported across six independently developed branches of established physics and formal science, is $G = (D, T, C)$, with retention R emerging as a theorem rather than a postulate. This is offered as a rigorous, falsifiable, and honestly incomplete research program — not as a closed case.

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